

Bangladesh University of Engineering and Technology

CSE208: Data Structure and Algorithms Sessional II

Online 3

Section: A1 + A2

Time: 35 Minutes

Electrifying the Kingdom

The Kingdom of Valoria has N cities. To bring the kingdom into the modern age, the King wants to provide electricity to every city. There are two ways to provide electricity to a city:

1. **Build a Power Station:** You can build a power station directly inside city i for a cost of P_i .
2. **Connect to the Grid:** You can run a transmission cable between city i and city j with cost C_{ij} .

Rules of Electricity:

- If city i has a power station, it has electricity.
- If city i is connected via cable to city j , and city j has electricity (either from a station or another cable connection), then city i also has electricity.

You need to minimize the total cost so that every city has access to electricity.

Input Format

- First line: Integer N (number of cities).
- Second line: N space-separated integers representing $P_1, P_2, \dots, P_N$ (cost to build a station in each city).
- Third line: Integer M (number of potential cables).
- Next M lines: Three integers u, v, w (cable between city u and v with cost w).

Output Format

Print a single integer: the minimum total cost to electrify all cities.

Constraints

- $1 \leq N \leq 2,000$
- $1 \leq M \leq N \times (N-1) / 2$
- $1 \leq P_i, w \leq 10^9$

Samples

Sample 1

Input	Output
3 10 20 30 2 1 2 5 2 3 5	20

*Explanation: Build station at City 1 (Cost 10). Connect City 1 to 2 (Cost 5).
Connect City 2 to 3 (Cost 5). Total: $10 + 5 + 5 = 20$.*

Sample 2

Input	Output
5 100 1 100 1 100 3 1 2 50 3 4 50 2 3 10	162

Sample 3

Input	Output
3 5 5 5 2 1 2 100 2 3 100	15